

MODELING AND IMPLEMENTATION OF THE ELECTRICITY MARKET MONITORING SYSTEM

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The purpose of the paper is to build an electricity market monitoring system, in which, the market is simulated to get the reasonable electricity price to represent the health of the market, hence provide a reference for the market regulator to take proper measure for interfere the market.

Due to the extreme importance of the power system, the operation of the electricity market has drawn much concern. The introduction of the electricity market aims to improve the efficiency, but meanwhile has brought many new problems, which cannot be found in the traditional monopoly system. Especially, in the economy aspect, power system has to face market risk other than the physical security problem. To find proper measures for market realization and adjustment, the authority should examine them carefully before their implementation. An electricity market monitoring system, which on one hand provides the references for market participants making strategic decisions and on the other hand is able to evaluate the impacts of specific measures or market context variation, is presented in this paper to help the regulator make decisions on market policy and the market participants foresee the market evolution. The monitoring system consists of two units, one is the market simulator, which detailed models the interaction between the market players by game theory and multi-agent system; the other is price forecast, which interprets the market evolution by mathematical or artificial intelligence approaches to foresee the prices. The two units cooperatively build a monitoring system for predicting the future market scenarios in different respects.

The economic impact of the China's energy saving policy is examined with the monitoring system, the corresponding financial cost is evaluated and the proper countermeasure in the respect of compensation policy for the power plants providing auxiliary service is proposed.

Keywords

Monitoring system, Market Simulator, Electricity market, Price Forecast, Bidding strategy